1 Angles & Trigonometry Basics

Angle $\theta = \dfrac{\text{arc}}{\text{radius}} = \dfrac{s}{r}$ (radian). $\pi\,\text{rad} = 180^\circ$, $1\,\text{rad} \approx 57.3^\circ$. Arc length $s = r\theta$.

Standard values

θ	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

Signs by quadrant (ASTC) & allied angles

- Q1 all $+$; Q2 $\sin, \csc$ $+$; Q3 $\tan, \cot$ $+$; Q4 $\cos, \sec$ $+$.
- $\sin(-\theta) = -\sin\theta$, $\cos(-\theta) = \cos\theta$, $\tan(-\theta) = -\tan\theta$.
- $\sin(90^\circ - \theta) = \cos\theta$, $\cos(90^\circ - \theta) = \sin\theta$.

2 Trigonometric Identities

- $\sin^2\theta + \cos^2\theta = 1$, $1 + \tan^2\theta = \sec^2\theta$, $1 + \cot^2\theta = \csc^2\theta$
- $\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$
- $\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$
- $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$
- $\sin 2\theta = 2\sin\theta\cos\theta$, $\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$
- $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$
- $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

3 Useful Approximations

For small θ (in **radians**) and $|x| \ll 1$:

- $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1 - \dfrac{\theta^2}{2}$
- Binomial: $(1+x)^n \approx 1 + nx$
- $\dfrac{1}{1+x} \approx 1 - x$, $\dfrac{1}{1-x} \approx 1 + x$, $\sqrt{1+x} \approx 1 + \dfrac{x}{2}$
- $e^x \approx 1 + x$, $\ln(1+x) \approx x$

> These drive error analysis and physics limits (pendulum, lens/mirror, fringe shifts). Always convert angles to radians first.

4 Differentiation — Standard Derivatives

$f(x)$	$\frac{d}{dx}f(x)$
c (constant)	0
x^n	$n\,x^{n-1}$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x \tan x$
$\csc x$	$-\csc x \cot x$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$

5 Rules of Differentiation

- Sum: $(u \pm v)' = u' \pm v';\ (cu)' = c\,u'$
- Product: $(uv)' = u'v + uv'$
- Quotient: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$
- Chain: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, i.e. $\frac{d}{dx} f\big(g(x)\big) = f'\big(g(x)\big)\, g'(x)$

6 Maxima & Minima

- At a maximum or minimum: $\frac{dy}{dx} = 0$.
- $\frac{d^2y}{dx^2} < 0 \Rightarrow$ maximum; $\frac{d^2y}{dx^2} > 0 \Rightarrow$ minimum.
- $\frac{d^2y}{dx^2} = 0 \Rightarrow$ test further (possible point of inflection).

7 Differentiation in Physics

- Velocity $v = \frac{dx}{dt}$; Acceleration $a = \frac{dv}{dt} = \frac{d^2x}{dt^2}$.
- $a = v\frac{dv}{dx}$ (useful when a depends on position).

- Angular: $\omega = \dfrac{d\theta}{dt}$, $\alpha = \dfrac{d\omega}{dt}$.
- Instantaneous rate = slope of the tangent to the graph.

8 Integration — Standard Integrals

$\int f(x)\,dx$	RESULT $(+C)$
$\int x^n\,dx\ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$
$\int \dfrac{1}{x}\,dx$	$\ln\lvert x\rvert$
$\int e^x\,dx$	e^x
$\int a^x\,dx$	$\dfrac{a^x}{\ln a}$
$\int \sin x\,dx$	$-\cos x$
$\int \cos x\,dx$	$\sin x$
$\int \sec^2 x\,dx$	$\tan x$
$\int \csc^2 x\,dx$	$-\cot x$
$\int \sec x \tan x\,dx$	$\sec x$
$\int \dfrac{dx}{1+x^2}$	$\tan^{-1} x$

9 Methods & Definite Integration

- $\int (u \pm v)\,dx = \int u\,dx \pm \int v\,dx$; $\int c\,f\,dx = c\int f\,dx$
- $\int f(ax+b)\,dx = \dfrac{1}{a}F(ax+b) + C$
- Definite: $\int_a^b f(x)\,dx = F(b) - F(a)$
- Properties: $\int_a^b = -\int_b^a$, $\int_a^a = 0$
- Area under a curve $= \int y\,dx$.

10 Integration in Physics

QUANTITY	INTEGRAL
Velocity from acceleration	$v = \int a\,dt$
Displacement from velocity	$\Delta x = \int v\,dt$
Work from force	$W = \int \vec{F} \cdot d\vec{x}$
Impulse from force	$\vec{J} = \int \vec{F}\,dt$
Charge from current	$q = \int i\,dt$

> Each is the **area under the corresponding graph** (a-t, v-t, F-x, ...).

11 Vectors — Basics & Addition

- **Scalar:** magnitude only. **Vector:** magnitude $+$ direction.
- Triangle / parallelogram law of vector addition.

Resultant of $\vec{A}$ and $\vec{B}$ at angle θ

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}, \qquad \tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

- $R_{\text{max}} = A + B\ (\theta = 0°)$; $R_{\text{min}} = |A - B|\ (\theta = 180°)$.
- Perpendicular: $R = \sqrt{A^2 + B^2}$.
- Subtraction: $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$, $|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$.

12 Vector Components & Unit Vectors

- $A_x = A\cos\theta,\ \ A_y = A\sin\theta$.
- $\vec{A} = A_x\,\hat{i} + A_y\,\hat{j} + A_z\,\hat{k}$.
- Magnitude $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$.
- Unit vector $\hat{A} = \dfrac{\vec{A}}{|\vec{A}|}$.
- $\hat{i}, \hat{j}, \hat{k}$ are mutually perpendicular unit vectors along x, y, z.

13 Dot (Scalar) Product

- $\vec{A} \cdot \vec{B} = AB\cos\theta$ (a **scalar**).
- Component form: $\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$.
- Commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$; $\vec{A} \cdot \vec{A} = A^2$.
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$; $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$.

- $\cos\theta = \dfrac{\vec{A}\cdot\vec{B}}{AB}$; projection of $\vec{A}$ on $\vec{B} = \dfrac{\vec{A}\cdot\vec{B}}{B}$.

Physics: Work $W = \vec{F}\cdot\vec{d}$; Power $P = \vec{F}\cdot\vec{v}$.

14 Cross (Vector) Product

- $\vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$ (a **vector** $\perp$ to both; right-hand rule).
- $|\vec{A}\times\vec{B}| = AB\sin\theta =$ area of the parallelogram.
- Anti-commutative: $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$; $\vec{A}\times\vec{A} = \vec{0}$.
- Cyclic: $\hat{i}\times\hat{j} = \hat{k},\ \ \hat{j}\times\hat{k} = \hat{i},\ \ \hat{k}\times\hat{i} = \hat{j}$.

Determinant form

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Physics: Torque $\vec{\tau} = \vec{r}\times\vec{F}$; Angular momentum $\vec{L} = \vec{r}\times\vec{p}$; $\vec{F} = q\,\vec{v}\times\vec{B}$.

15 Graphs & Standard Curves

CURVE	EQUATION
Straight line	$y = mx + c$, slope $m = \tan\theta = \dfrac{\Delta y}{\Delta x}$
Line through origin	$y = mx$
Parabola	$y = kx^2$ or $y^2 = kx$
Rectangular hyperbola	$xy = c$ i.e. $y = \dfrac{c}{x}$
Exponential decay	$y = y_0\,e^{-kx}$
Circle	$x^2 + y^2 = a^2$
Sine wave	$y = A\sin(\omega x)$

16 Algebra Toolkit

Quadratic $ax^2 + bx + c = 0$

- $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; discriminant $D = b^2 - 4ac$.
- $D > 0$: two real; $D = 0$: equal; $D < 0$: complex.
- Sum of roots $= -\dfrac{b}{a}$; product $= \dfrac{c}{a}$.

Logarithm & exponents

- $\log(mn) = \log m + \log n,\quad \log\dfrac{m}{n} = \log m - \log n,\quad \log m^n = n\log m$
- $a^m a^n = a^{m+n},\ \ (a^m)^n = a^{mn}$

Componendo & dividendo: if $\dfrac{a}{b} = \dfrac{c}{d}$ then $\dfrac{a+b}{a-b} = \dfrac{c+d}{c-d}$.

17 Quick Recall / Physics Tips

- Calculus & approximations need angles in **radians**.
- Small θ: $\sin\theta \approx \tan\theta \approx \theta, \quad \cos\theta \approx 1$.
- x–t slope $= v$; v–t slope $= a$; area under v–$t = \Delta x$; area under a–$t = \Delta v$; area under F–$x = W$.
- Dot product $\rightarrow$ scalar (work, power); cross product $\rightarrow$ vector (torque, $\vec{L}$).
- $(1+x)^n \approx 1 + nx$ for $|x| \ll 1$.
- Perpendicular vectors: $R = \sqrt{A^2 + B^2}$.